

Imperial College London

BSc/MSci EXAMINATION June 2024

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Monday, 3rd June 2024: 14:00 to 16:00

*The paper consists of **three** questions – please answer **all three**.*

A table of Laplace transforms is included at the end of the paper.

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 3 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 3 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. Consider a low-pass filter implemented through a simple RC circuit.

- (i) Show that the complex voltage gain for a sinusoidal input at angular frequency $\omega = 2\pi f$ is given by

$$\mathbf{G} = \frac{1 - j\omega RC}{1 + \omega^2 R^2 C^2}.$$

[5 marks]

- (ii) Derive and sketch the Bode plot for the filter, annotating key features.

[5 marks]

- (iii) Propose, with a brief justification, values for the R and C components of the filter serving each of these cases; comment in each case why a filter is needed.

- (a) A 1 MS/s digitiser with high input impedance.
- (b) A phase detector based on a synchronous rectifier driven at 100 Hz.
- (c) An audio amplifier driving a 'subwoofer' speaker below 100 Hz.

[10 marks]

- (iv) Consider an RC filter with components R and C driving a resistive load R . Derive the complex gain of the system, and hence explain how this loading affects the frequency response over the whole spectrum.

[8 marks]

- (v) The output of a sensor producing a Gaussian response with unit amplitude and FWHM $\Delta T = 10 \mu\text{s}$ is filtered by a low-pass circuit with $RC = \Delta T$ and then digitised at 100 kS/s with a high-resolution ADC; you may ignore loading effects. Sketch the frequency response of the sensor before and after the filter, and superpose that of the digitised signal; you may use arbitrary units for the y axis. A useful Fourier pair is given below.

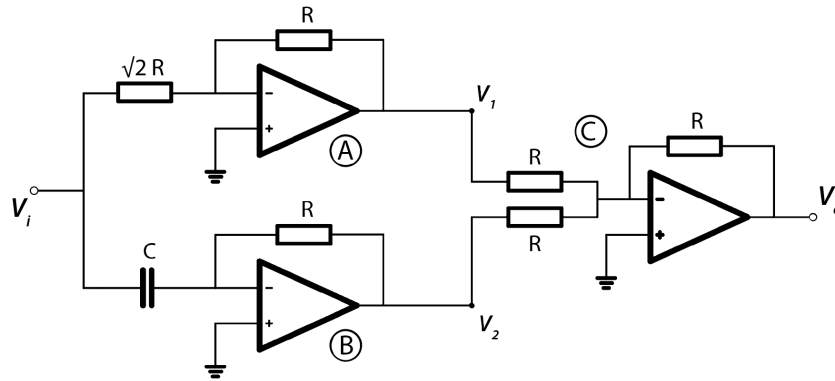
[12 marks]

[Total 40 marks]

You may need the following Fourier transform:

$$\mathcal{F} \left\{ \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{t^2}{2\sigma^2}\right) \right\} = \exp\left(-\frac{\sigma^2\omega^2}{2}\right).$$

2. (i) The circuit shown in the figure comprises three functional blocks: A, B and C.



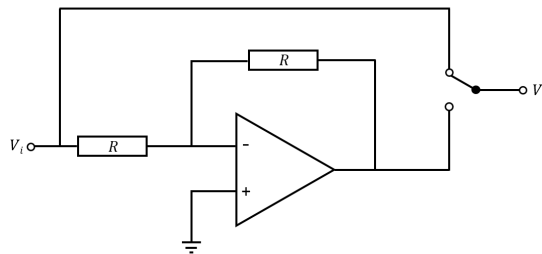
Give the circuit analysis and hence derive the input/output relationship for each of the following, justifying your assumptions:

- For block A, find the relation between V_1 and V_i .
- For block B, find the relation between V_2 and V_i .
- For block C, find the relation between V_1 , V_2 and V_o .
- For an input $V_i = \sin \omega t$ and for $\omega = (\sqrt{2}RC)^{-1}$, show that the output is

$$V_o = \sin\left(\omega t + \frac{\pi}{4}\right).$$

[15 marks]

- (ii) (a) In the circuit below the output switch is controlled by a square-wave signal of angular frequency ω . Name this circuit and describe its behaviour.



- For the case that the input signal is $V_i = A \sin(\omega t + \phi)$, and with zero phase ($\phi = 0$), make a sketch of the input, the reference and the output signal V_o .
- What is the main advantage of this circuit compared with the traditional circuit based on a four-diode bridge?
- Explain how to adapt the circuit in order to perform an averaging function on V_o . What is the name given to this new circuit?
- Show that the output of the new circuit is

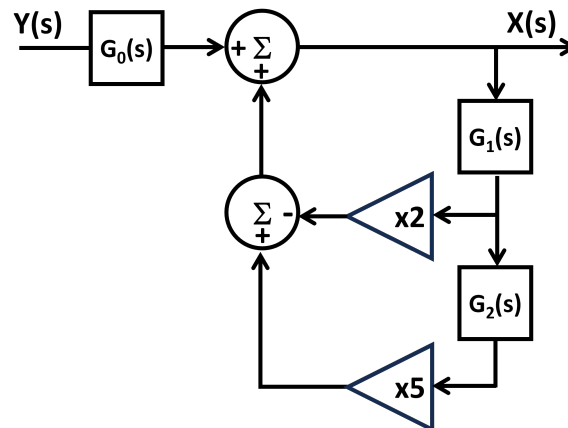
$$\langle V \rangle = \frac{2A}{\pi} \cos \phi.$$

[15 marks]

[Total 30 marks]

You may need the identity $\sin(a + b) = \sin a \cos b + \sin b \cos a$.

3. The system shown below contains three functional blocks $G_{0..2}$ plus two ideal amplifiers of gain 2 and 5, and features both positive and negative feedback.



- (i) (a) Calculate the system's transfer function in terms of $G_0(s)$, $G_1(s)$ and $G_2(s)$.
- (b) Assuming $G_0(s)$ has a time-domain impulse response of $8t e^{-2t}$, find its s -domain representation.
- (c) Assuming $G_1(s)$ is the ideal differentiator with unit gain, find its s -domain representation.
- (d) Sketch the circuit which would have the transfer function $G_1(s)$ assuming it contains two ideal operational amplifiers.
- (e) Assuming $G_2(s)$ is the ideal integrator with unit gain, find its s -domain representation.
- (f) Find the overall transfer function in terms of s .

[18 marks]

- (ii) (a) Is the system stable? Explain your answer.
- (b) Would it oscillate? Explain why.
- (c) Determine the system response to the unit step input in the time-domain.

[12 marks]

[Total 30 marks]

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Figure 1: Laplace transforms.